Cmod A7-35T Internal IP Peripheral Reference

Platform: Xilinx Artix-7 (xc7a35tcpg236-1) — Cmod A7-35T
Processor: MicroBlaze RISC-V
System Clock: 100 MHz (12 MHz on-board oscillator × PLL)
Toolchain: Vivado & Vitis 2025.2

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1. Processor Core & System Infrastructure

1.1 MicroBlaze RISC-V (`microblaze_riscv_0`)

Item	Value
Architecture	32-bit RISC-V — RV32IM + bit-manipulation (hardware multiply/divide)
Clock	100 MHz
Buses	LMB to local BRAM (1-cycle) · AXI to peripherals and external memory
Cache	16 KB I-Cache + 16 KB D-Cache, write-through, 32 B lines — covers exactly the 512 KB SRAM (`0x6000_0000` – `0x6007_FFFF`)
Debug	JTAG: breakpoints, register inspection, memory read/write

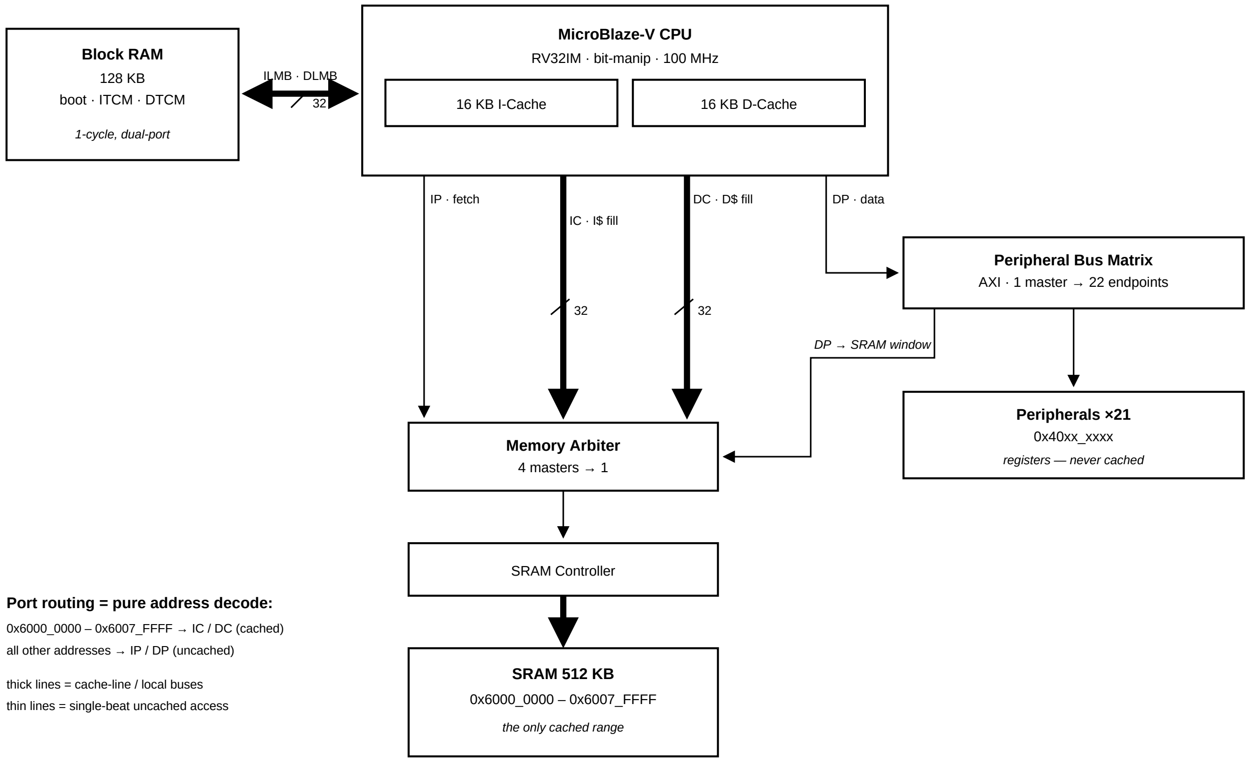
Description: The main processor core. Executes user firmware and reaches every peripheral through the AXI interconnect. Only the SRAM range is cached — BRAM is already single-cycle, and peripheral registers must never be cached.

1.2 Local Memory

Item	Value
Address	`0x0000_0000` – `0x0001_FFFF` (128 KB)
Access	1 cycle, dual-port — instruction and data sides read simultaneously
Layout	32 KB bootloader + 32 KB ITCM + 64 KB DTCM

Description: Instruction and data local memory implemented with FPGA Block RAM — functionally this MCU's tightly-coupled memory (TCM): the ILMB/DLMB ports play the same role as ITCM/DTCM on other MCU cores (e.g. Cortex-M7), giving 1-cycle access outside the cache path. Layout: `0x0000` – `0x7FFF` (32 KB) holds the UART bootloader (restored from flash at every configuration); `0x8000` – `0xFFFF` (32 KB) is the application's ITCM for interrupt handlers and timing-critical code (`ITCM_FUNC` , copied out of the SRAM image by `mcu_init()` at startup); `0x10000` – `0x1FFFF` (64 KB) is the DTCM, holding the stack (top-down from `0x20000`) and `DTCM_DATA` fast data.

1.3 AXI SmartConnect (`microblaze_riscv_0_axi_periph`)



Description: AXI crossbar that fans the processor's data port out to all peripheral endpoints. Routing is pure address decoding — the diagram above shows which path each address range takes (a second, smaller interconnect merges the cache and debug masters in front of the SRAM controller).

1.4 AXI Interrupt Controller (`microblaze_riscv_0_axi_intc`)

Item	Value
Base Address	`0x4040_0000`
Interrupt Sources	8

Interrupt Mapping:

Channel	Source	Description
In0	`timer_0`	System timer 0 interrupt
In1	`timer_1`	System timer 1 interrupt
In2	`timer_2`	System timer 2 interrupt
In3	`uart_1`	External UART interrupt
In4	`uart_USB`	USB UART interrupt
In5	`INT_0_3`	External GPIO interrupt (4-bit)
In6	`i2c_0`	I2C controller interrupt

Channel	Source	Description
In7	`spi_0`	External SPI master interrupt

1.5 Debug Module (`mdm_1`)

Description: MicroBlaze Debug Module providing JTAG debug access for breakpoints, register inspection, and memory read/write. It can also trigger a system reset from the debugger (used by `xsdb` / Vitis debug sessions).

1.6 Clocking Wizard (`clk_wiz_1`)

Description: Multiplies the 12 MHz on-board oscillator up to the 100 MHz system clock with an MMCM/PLL. Its `locked` signal indicates clock stability and gates the release of system reset.

1.7 Processor System Reset (`rst_clk_wiz_1_100M`)

Description: Generates synchronized reset signals for the CPU, bus fabric and peripherals, releasing them only after the clock is stable. External reset source: on-board button BTN0 (A18, active-high).

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2. UART Communication

2.1 USB UART (`uart_USB`)

Item	Value
Base Address	`0x4030_0000`
TX / RX Pins	J18 / J17 — routed to the on-board Micro-USB connector
Interrupt	INTC In4

Description: 16550-compatible UART for host PC communication through the on-board Micro-USB connector. Baud rate is software-configured via the divisor latch: at 100 MHz system clock, divisor 54 (0x36) gives 115200 baud. Typically mapped as STDIN/STDOUT.

2.2 External UART (`uart_1`)

Item	Value
Base Address	`0x4031_0000`
TX / RX Pins	J1 (DIP 11) / K2 (DIP 12)
Interrupt	INTC In3

Description: Second 16550 UART exposed on the DIP connector for communication with external devices (e.g., sensor modules, Bluetooth modules).

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3. GPIO (General Purpose I/O)

All GPIO groups are memory-mapped ports with per-bit direction control: the TRI register sets each pin's direction, the DATA register reads/writes the pin. Vitis driver: `XGpio`.

3.1 On-Board GPIO

Instance	Base Address	Width	Direction	Connection	Description
`board_led_2bits`	`0x4000_0000`	2	Output	A17, C16	On-board LEDs × 2
`board_button`	`0x4001_0000`	1	Input	B18	On-board user button (BTN1)
`board_rgb`	`0x4002_0000`	3	Output	B17, B16, C17	On-board RGB LED (R/G/B)

3.2 DIP Connector GPIO (4 Groups × 7-bit)

Instance	Base Address	Width	DIP Pins	Description
`gpio_A_0_6`	`0x4003_0000`	7	Pin 1–7	GPIO Group A, bidirectional I/O
`gpio_B_0_6`	`0x4004_0000`	7	Pin 17–23	GPIO Group B, bidirectional I/O
`gpio_C_0_6`	`0x4005_0000`	7	Pin 42–48	GPIO Group C, bidirectional I/O
`gpio_D_0_6`	`0x4006_0000`	7	Pin 26–32	GPIO Group D, bidirectional I/O

Description: Each group is a 7-bit bidirectional port; every bit can be an input or an output independently.

3.3 External Interrupt Inputs (`INT_0_3`)

Item	Value
Base Address	`0x4007_0000`
Width	4-bit, input-only
DIP Pins	Pin 8 (INTR_0), Pin 9 (INTR_1), Pin 41 (INTR_2), Pin 33 (INTR_3)
Interrupt	INTC In5

Description: Four external interrupt inputs grouped into a single GPIO instance with interrupt generation enabled — a change on any of the four pins can raise INTC In5.

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4. Timers & PWM

Six 32-bit AXI timer instances (Vitis driver: `XTmrCtr`) — three as general-purpose timers with interrupts, three with their outputs routed to pins as PWM channels.

4.1 System Timers

Instance	Base Address	Interrupt	Description

Instance	Base Address	Interrupt	Description
`timer_0`	`0x4010_0000`	INTC In0	General-purpose system timer
`timer_1`	`0x4011_0000`	INTC In1	General-purpose timer
`timer_2`	`0x4012_0000`	INTC In2	General-purpose timer

4.2 PWM Outputs

Instance	Base Address	Output Pin	DIP Pin	Description
`PWM_0`	`0x4020_0000`	J3	Pin 10	PWM Channel 0
`PWM_1`	`0x4021_0000`	W3	Pin 34	PWM Channel 1
`PWM_2`	`0x4022_0000`	W4	Pin 40	PWM Channel 2

Description: Timer instances configured in PWM mode to generate square-wave outputs. Useful for LED dimming, motor speed control, buzzer tone generation, etc. Frequency and duty cycle are configured through the Timer Load Registers and the PWM enable bit.

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5. External Memory

5.1 QSPI Flash (`axi_quad_spi_0`)

Item	Value
Base Address	`0x4050_0000`
Flash Device	On-board 4 MB QSPI NOR flash (Macronix; Micron on older boards)
Mode	CPU-programmable register mode — flash contents are **not** memory-mapped
FIFO	256 B TX/RX — one full flash page (256 B) per transfer

Description: Controls the on-board Quad-SPI NOR Flash. The full register set (control, status, TX/RX FIFO, slave select) is exposed at `0x4050_0000` , so the CPU can issue any SPI command — read (`0x0B`), Write Enable (`0x06`), Sector/Block Erase (`0x20` / `0xD8`), Page Program (`0x02`), status poll (`0x05`). This is what allows the UART bootloader to program application images into flash at runtime (`workspace-example/bootloader/src/bootloader.c` is a complete worked example, and the Vitis `XSpi` driver wraps the register protocol).

Flash contents are **not memory-mapped**: there is no XIP window, and code cannot execute from flash directly. The standalone-boot design instead copies the application from flash into SRAM at power-on (see the Standalone Boot Mode guide).

> **Design note:** a read-only memory-mapped XIP window and CPU-programmable register mode are mutually exclusive in this controller. This project chooses register mode: self-programming (Arduino-style `upload.py` workflow) was judged more valuable for the course than execute-in-place, and the 512 KB SRAM + I-cache serves execution instead.

5.2 SRAM / Cellular RAM (`axi_emc_0`)

Item	Value
Base Address	`0x6000_0000`
Size	512 KB — `0x6000_0000` – `0x6007_FFFF`, an exact physical fit
Memory	On-board asynchronous SRAM (Cellular RAM)
Cache	Fully covered by the I-Cache and D-Cache

Description: External memory controller for the on-board 512 KB SRAM — the main application memory: standalone boot copies the program here and executes it. Raw access latency is higher than Block RAM, but with both caches covering this range, hot code and data run near BRAM speed.

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6. Analog-to-Digital Conversion (XADC)

6.1 XADC Wizard (`xadc_wiz_0`)

Item	Value
Base Address	`0x4060_0000`
Resolution	12-bit
Conversion Rate	500 KSPS aggregate
Sequencer	Continuous scan over the 5 enabled channels

Enabled Channels:

Channel	Source	Description
VAUX4	DIP Pin 15 (G3/G2)	External analog input 0 (on-board divider: 0–3.3 V → 0–1 V)
VAUX12	DIP Pin 16 (H2/J2)	External analog input 1 (on-board divider: 0–3.3 V → 0–1 V)
Temperature	Internal	FPGA die temperature monitor
VCCINT	Internal	Core voltage monitor (1.0 V)
VCCAUX	Internal	Auxiliary voltage monitor (1.8 V)

Description: The Artix-7 built-in 12-bit ADC capable of measuring external analog signals and monitoring internal FPGA temperature and supply voltages. External input pins pass through an on-board resistive voltage divider (2.32 KΩ / 1 KΩ, ratio ≈ 0.301), accepting up to 3.3 V at the DIP pin.

Effective Per-Channel Sampling Rate: The sequencer continuously cycles through all 5 enabled channels (VAUX4, VAUX12, Temperature, VCCINT, VCCAUX). The aggregate conversion rate is 500 KSPS, giving each channel an effective rate of 500 K ÷ 5 = **100 KSPS**.

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7. Serial Expansion Interfaces (I2C / SPI)

7.1 I2C Controller (`i2c_0`)

Item	Value
Base Address	`0x4070_0000`
SCL Frequency	100 kHz (standard mode)
Input Glitch Filter	50 ns on SCL and SDA — per the I2C tSP spike-suppression spec
SCL / SDA Pins	DIP Pin 13 (L1) / Pin 14 (L2)
Pull-ups	Weak FPGA internal pull-ups enabled; **external 4.7 kΩ to 3.3 V recommended** for real devices
Interrupt	INTC In6

Description: AXI IIC master for external I2C devices (sensors, EEPROMs, OLED displays). Open-drain signaling: any device may only pull the line low; the pull-up resistor returns it high. Devices are addressed by their 7-bit I2C address. Use the Vitis `XIic` driver.

7.2 External SPI Master (`spi_0`)

Item	Value
Base Address	`0x4080_0000`
SCLK	Software-programmable, ≈1.6 – 25 MHz (**6.25 MHz at reset**)
Clock Control	`0x4090_0000` — runtime clock setting; see `examples/07_spi_clock` for the `spi_set_clock()` helper
Slave Selects	2 — two devices can share the bus
Pins	DIP 35 SCLK (V3) · 36 MOSI (W5) · 37 MISO (V4) · 38 SS0 (U4) · 39 SS1 (V5)
Interrupt	INTC In7

Description: SPI master for external devices (displays, ADCs, flash modules). Full-duplex push-pull signaling — no pull-ups needed; the active device is chosen by driving its SS line low. Use the Vitis `XSpi` driver. The serial clock is set at runtime through the clock-control block: slow it down for breadboard wiring or long cables, speed it up for short-wired display modules (`spi_set_clock(hz)` — one call, takes effect immediately).

> **Note:** This is a *second, independent* SPI controller — do not confuse it with `axi_quad_spi_0` (`0x4050_0000`), which is dedicated to the on-board QSPI boot flash. Firmware should select controllers by instance macro (`XPAR_SPI_0_BASEADDR` vs `XPAR_AXI_QUAD_SPI_0_BASEADDR`), never by generic `XPAR_XSPI_n_*` numbering.

8. Complete Address Map

Peripheral addresses follow a class-based convention — `0x40[C]x_xxxx`**, where** `C` **is the peripheral class** (1 MB per class, 64 KB per instance). Reading an address immediately tells you what kind of device it is: class 0 = GPIO, 1 = Timer, 2 = PWM, 3 = UART, 4 = INTC, 5 = QSPI control, 6 = XADC, 7 = I2C, 8 = SPI, 9 = SPI clock control.

Base Address	Range	Peripheral	Type	Category
`0x0000_0000`	128K / 128K	Local Memory (BRAM)	BRAM	Memory
`0x4000_0000`	64K	board_led_2bits	axi_gpio	GPIO
`0x4001_0000`	64K	board_button	axi_gpio	GPIO
`0x4002_0000`	64K	board_rgb	axi_gpio	GPIO
`0x4003_0000`	64K	gpio_A_0_6	axi_gpio	GPIO
`0x4004_0000`	64K	gpio_B_0_6	axi_gpio	GPIO
`0x4005_0000`	64K	gpio_C_0_6	axi_gpio	GPIO
`0x4006_0000`	64K	gpio_D_0_6	axi_gpio	GPIO
`0x4007_0000`	64K	INT_0_3	axi_gpio	Interrupt
`0x4010_0000`	64K	timer_0	axi_timer	Timer
`0x4011_0000`	64K	timer_1	axi_timer	Timer
`0x4012_0000`	64K	timer_2	axi_timer	Timer
`0x4020_0000`	64K	PWM_0	axi_timer	PWM
`0x4021_0000`	64K	PWM_1	axi_timer	PWM
`0x4022_0000`	64K	PWM_2	axi_timer	PWM
`0x4030_0000`	64K	uart_USB	axi_uart16550	Communication
`0x4031_0000`	64K	uart_1	axi_uart16550	Communication
`0x4040_0000`	64K	axi_intc	axi_intc	System
`0x4050_0000`	64K	axi_quad_spi_0 (QSPI flash controller)	axi_quad_spi	Memory
`0x4060_0000`	64K	xadc_wiz_0	xadc_wiz	ADC
`0x4070_0000`	64K	i2c_0 (DIP 13/14)	axi_iic	Communication
`0x4080_0000`	64K	spi_0 (external master, DIP 35–39)	axi_quad_spi	Communication
`0x4090_0000`	64K	spi_0_clk (SPI clock control)	clock control	Communication

Base Address	Range	Peripheral	Type	Category
`0x6000_0000`	512K	axi_emc_0 (exact physical fit, I/D-cached)	axi_emc	Memory